(a) Correction Inertia — "He felt very purple."

T2T Editing

Step t: He | felt | very | purple | .

$\max_v p_\theta(v) = 0.12 < \tau_{t2t} = 0.5$

"sad": 0.12 · "happy": 0.11 · "excited": 0.05

→ No action (multimodal — no candidate qualifies)

Step t+1: He | felt | very | purple | .

✗ *stuck — error persists, pollutes context*

Correction inertia

Error detected but cannot be corrected — no single replacement exceeds threshold

T2M Remasking

Step t: He | felt | very | purple | .

$p_\theta(\text{"purple"}) = 0.00002 < \tau_{lp} = 0.5$

→ Reset to [M] (remask)

Step t+1: He | felt | very | [M] | .

neutral signal — no semantic bias

↓ M2T re-predicts under clean context

Step t+2: He | felt | very | sad | .

✓ *correct — context purification succeeds*

Context purification + Delayed commitment

Clean context enables correct re-prediction

(b) Premature Replacement under Incomplete Context — DROP: "How many are military specialists or civil servants?" (gold = 857)

T2T Editing → "657" (wrong)

Block i — answer digits

Step t: The | answer | is | 8 | 5 | [M]

$p(8)=0.11, p(6)=0.64 > \tau_{t2t}=0.5$ → replaces 8→6

Step t+1: The | answer | is | 6 | 5 | 7

✗ *"657" committed with error at hundreds digit*

Block i+1 — continuation (sees "657" as frozen context)

Block i+1: … | 657 | is | wrong | …

Erroneous "657" is frozen → pollutes all subsequent predictions

Premature replacement

At t=1 neighbors "5" and "7" are still [M] → context incomplete. T2T replaces "8"→"6" based on unreliable signal. Once committed, irreversible.

T2M Remasking → "857" (correct)

Block i — answer digits

Step t: The | answer | is | 8 | 5 | [M]

$p(8)=0.11 < \tau_{lp}=0.5$ → remask 8→[M]

Step t+1: The | answer | is | [M] | 5 | 7

↓ M2T re-predicts: $p(8)=0.94 > \tau_{m2t}=0.7$ → fill

Step t+2: The | answer | is | 8 | 5 | 7

✓ *"857" → clean context for block i+1*

Context purification + Delayed commitment

T2M remasks "8" instead of replacing it. At t=2, "5" and "7" are committed → richer context → re-predicts "8" with p=0.94.

(c) Delayed Commitment — DROP: "Who threw the first touchdown pass?" (gold = Jon Kitna)

T2T Editing → "Kitna" (wrong)

t=1: The | answer | is | Jon | na | [M] | [M]

M2T fills "Jon"(0.82) "na"(0.74)

t=2–3: The | answer | is | Jon | na | . | <end>

"Jon"(p=0.59→0.30) drops as more context fills in

t=4: t=4: p("Jon")=0.30, top1="Kit" p=0.57 > τ_{t2t}=0.5

→ T2T replaces "Jon" → "Kit"

The | answer | is | Kit | na | . | <end>

✗ *"Kitna" — "Jon" irrecoverably replaced*

Premature commitment

p("Jon")=0.30 while p("Kit")=0.57 > τ → T2T replaces "Jon"→"Kit". "Kit"+"na" reinforces "Kitna" — "Jon" is irrecoverable.

T2M Remasking → "Jon Kitna" (correct)

t=1: The | answer | is | Jon | na | [M] | [M]

Same start: "Jon"(0.82) "na"(0.74)

t=2: p("na")=0.21 < τ_{lp}=0.5 → remask; "."(0.82) fills

t=2: … | Jon | [M] | . | [M]

t=3: [M]→"Kit"(0.91) fill; p(".")=0.45 → remask

t=4: p("Kit")=0.47 → remask; [M]→"na"(0.96) fill

t=5–6: "Kit"(1.00) re-fill; "na"(0.99) re-fill

t=7: The | answer | is | Jon | Kit | na | .

✓ *All p≥0.98 → "Jon Kitna"* ✓

Delayed commitment

"Kit" and "na" remask 6 times total; each re-prediction sees more converged context. "Jon" preserved (p≥0.69); final: "Kit"(0.98)+"na"(0.99).

Correct | Erroneous | T2T edited | T2M remasked | Mask [M] | Re-predicted